Sparse demand needs quantile forecasts, but few studies compare these models leak-safely

Problem

Orders need quantiles, not means: $q^* = Cu / (Cu + Co)$

59% zeros; 90% of series intermittent or lumpy

Censoring: zero runs may be stockouts; M5 has no inventory records

Existing approaches

Intermittent [2–4] Croston, SBA, TSB: point forecasts, no covariates; cited, not run

State space [9–11] Level + seasonal states; missing days skip the update, so masking is easy

LightGBM [12, 13] Fast boosting; widely used among top M5 accuracy entries

DeepAR [16] Global autoregressive LSTM; probabilistic forecasts by sampling

Evaluation [1, 13] Pinball loss is a proper score; simple seasonal benchmarks are hard to beat

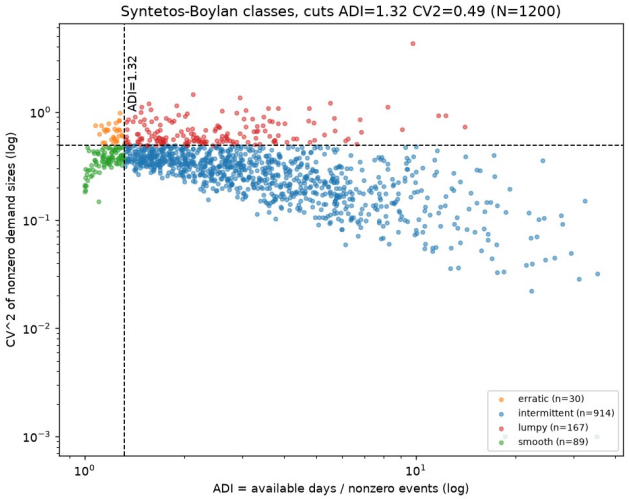
Objective

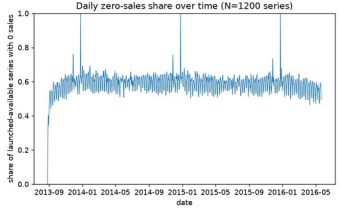
How do a local state-space model, a tabular GBDT, a hybrid and a global deep model compare on accuracy, calibration and cost under one leak-safe protocol?

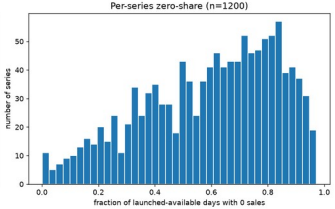
Gap: few studies compare all three families with censoring and calibration treated consistently. A careful comparison, not a new method.

References (numbers as in the paper): [1] Gneiting & Raftery 2007 · [2–4] Croston 1972; Syntetos & Boylan 2005; Teunter et al. 2011 · [9–11] Kalman 1960; Harvey 1989; Durbin & Koopman 2012 · [12] Ke et al. 2017 · [13] Makridakis et al. 2022 · [16] Salinas et al. 2020.

1,200 clean but sparse series: 59% zeros, mostly intermittent, and no verified stockout labels







59.11%
zero sales, launched and available cells

914 intermittent

167 lumpy

89 smooth

30 erratic

Data quality

- 3 stores × 400 items; 1,000 days
- No duplicates, negatives or missing sales; 701,313 price rows clean
- 95.64% of cells available

Patterns

- Weekends 1.26–1.31× category mean
- SNAP days +7.0 / +10.7 / +5.8% (CA/TX/WI)
- Event days 0.934× non-event
- 56.75% of series change price
- Descriptive, not causal

Stockout proxy

- Zero run L ≥ max(28, 3·ADI), ≥10 sales
- Flags 12.97% of cells (8.69% of test cells)
- Heuristic: M5 has no inventory records

Full-window descriptive statistics (including test days), not model inputs; the backtest recomputes classes before the first origin. ADI cut-off 1.32, CV^2 0.49. Associations are not causal. Source: report Section III, Figs. 1–2.

59 cutoff-aware features: history is read at the anchor day, known-future covariates at the target day

31 Target history read at anchor t, data up to t only

Lags y_t, y_{t-1}, y_{t-2} · weekday lags (4) + mean · rolling mean, sd, nonzero rate, masked share over 7/14/28/56/112 d · days since sale, age, mean to date

23 Known-future covariates read at target day t + h

Calendar (13) · events (5) · SNAP (1) · price (4): log price, weekly change, ratio to 26-week past mean, promo flag

5 Horizon and static Horizon h · department, category, store, state

+3 Kalman state features hybrid only, filter on last 514 days

Filtered level · one-step predictive sd · h-step state-space mean

Feature vector

59

features (hybrid: 62)

Rows: anchors 7 d apart over the last 364 d, with t + h ≤ T0

LightGBM

Direct multi-horizon: 4 buckets of 7 days, Poisson mean + 7 quantile models each

Price assumptions: target-week price is treated as known in advance (M5 convention). Change and past-mean features use earlier weeks only. The whole-window mean log-price fill for weeks without a posted price is not strictly causal; those cells are never scored.

Source: report Section IV, Table I. Features shown are those implemented; item_id is not a LightGBM feature in the main run. Counts: 31 + 23 + 5 = 59; the hybrid adds 3.

Four formulations, one rolling-origin design, and a leakage test that is evidence, not proof

Kalman filter

$$\sqrt{y_t} = \ell_t + \gamma_t + \varepsilon_t$$

Local level + weekly seasonal states; masked days skip the update; quantiles back-transformed (x^2).

Direct LightGBM

$$\hat{q}_{t+h} = f_q,b(h)(x_t)$$

4 horizon buckets, Poisson + 7 quantile models each; no recursion, no compounding.

Kalman–LightGBM

$$x_t + [\hat{\ell}, sd_1, Z\,T^h\,a_t]$$

Same LightGBM with three Kalman state features appended.

DeepAR

$$y_t \sim NegBin(\mu_t, \alpha_t)$$

Global 2-layer LSTM; μ from softplus of hidden state; 200 sampled paths.

Design and metrics

4 origins × 28 d; cells: 124,992 all / 114,127 non-censored / 9,408 cold start

WAPE = $\Sigma|y - \hat{y}^{0.5}| \,/\, \Sigma y$; MAE, RMSE

Pinball ρq(u) = max(qu, (q−1)u), mean of 7 levels

Cov80/90: share of y inside the interval

Leakage test: what it shows

Post-origin targets replaced by random integers: contexts unchanged, six model families agree within 10^{-5}, features and flags unchanged.

Strong evidence, not proof: one perturbation on a small fixture; misses e.g. the price fill.

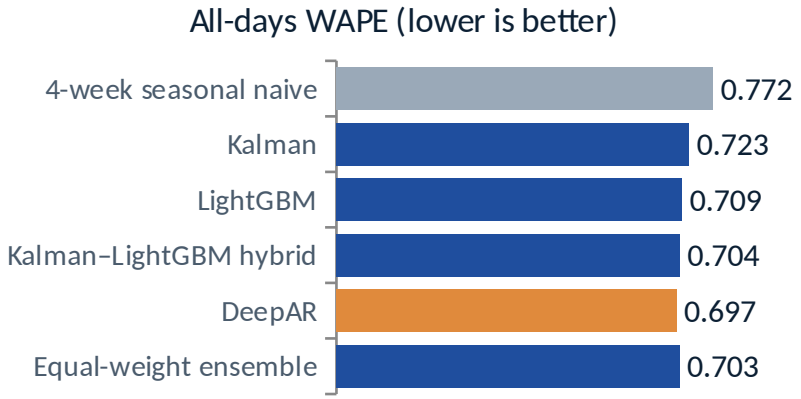
Calibration caveat

Inclusive bounds count y = ŷ = 0 as covered, so zero ties inflate coverage on zero-heavy counts.

Discrete-aware check: P(y<ŷ) ≤ q ≤ P(y≤ŷ), tolerance 0.02.

Source: report Sections IV–VI and VIII. u = y − ŷ^(q). Cells = (origin, series, horizon). The leakage test uses a small fixture and specific perturbations.

DeepAR wins all-days WAPE, the hybrid wins off flagged days, pinball is a tie



Model	All-days WAPE	All-days pinball	Non-cens. WAPE	Cold start WAPE
Seasonal naive (4 wk)	0.772	0.2899	0.759	0.875
Kalman	0.723	0.2736	0.686	0.839
LightGBM	0.709	0.2637	0.663	0.773
Kalman–LightGBM	0.704	0.2620	**0.660**	**0.760**
DeepAR	**0.697**	**0.2619**	0.665	0.773
Equal-weight ensemble	0.703	0.2623	0.665	0.783

Ablations run: 7 Kalman configs, 2 DeepAR variants

- Negative binomial → Gaussian: DeepAR WAPE 0.697 → 0.792
- Masking off: better all-days, worse non-censored (Kalman pinball 0.2907 vs 0.2822)

Not run: LightGBM and stockout-rule ablations, other DeepAR variants, tuning, seeds

Probabilistic: tied

Top-three pinball spread ≈ 0.0004 vs ≈ 0.004 across origins. No decisive winner.

DeepAR Cov90 = 0.965

Zero ties inflate inclusive coverage. Discrete-aware check passes at all 7 levels; not proof of calibration.

Lower is better. Teal = best in column; blue = top-three pinball, effectively tied (exact: hybrid 0.26198, DeepAR 0.26191, ensemble 0.26231, LightGBM 0.26366). Non-censored relies on the heuristic proxy; cold start is simulated. Tables II–IV.

The best model depends on the question you ask, and on the assumptions behind the answer

1 **0.697**

DeepAR: best all-days WAPE

Lowest MAE too (0.891); −9.7% WAPE vs seasonal naive

2 **0.660**

Hybrid: best off flagged days

Non-censored WAPE 0.65981; cold start 0.760 vs 0.773

3 **≈ tied**

Top three on pinball loss

0.26191 to 0.26231, vs ≈ 0.004 variation across origins

Practical implications

- Score in several settings; ranks can change
- Kalman ≈ MA28 accuracy, ≈ 29× faster than LightGBM
- Check proxy and price assumptions first

Main limitations

- One frozen M5 subset; four test origins
- Single seed; no hyperparameter tuning
- Heuristic stockout proxy; no inventory labels
- Simulated, not genuine, cold start
- Price-availability assumption; non-causal gap fill
- Croston, SBA, TSB, ETS, ARIMA: literature only

FUTURE WORK *not yet run*

- Hyperparameter tuning
- Multiple seeds
- LightGBM and stockout-rule ablations
- Remaining DeepAR variants
- Feature importance
- Croston, SBA, TSB baselines

Model choice depends on the evaluation objective and the validity of the underlying assumptions.

Source: report Sections VII–IX. Future work is listed for transparency; none of it has been run.